Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/30/2005
PI Name:	Tang, Jay X Ph.D.		
Project Title:	Microscopic Studies of Gravi-Sensitive Microtubule Assembly in Simulated Variable Gravity Conditions		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Cell Culture		
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Zip Code:	02912-1843	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-03
Start Date:	06/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Valles, Jr., James M (Brown University)		
Grant/Contract No.:	NNA04CC57G		
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Performance Goal Text:			

Dynamic assembly of tubulin is an essential biological process of fundamental importance to cell morphology, division and growth. We are investigating the mechanism of pattern formation during microtubule assembly, with the goal of understanding why polymerized microtubules form striated patterns under normal gravity but not under microgravity. A quantitative microscopic picture of the evolution of the patterns is acquired in the first phase of the program. This detailed characterization is used to test physical models of the process. The second phase of the study is to test how small perturbations, such as altered gravity, affect the pattern formation process. We will employ a novel Variable Magnetic Force (VMF) apparatus, the capabilities of which are such that the forces that couple to a solution's density fluctuations can be tuned to vary from microgravity to hypergravity. The threshold for the gravitational response of pattern formation for both polymerizing microtubules and microtubule asters will be investigated using this instrument. By examining how sensitive these patterns are to altered gravity, new insight into the effects of gravity in determining the biologically

Task Description:	relevant structures will be obtained. The results of the program may also lead to effective countermeasures to the adverse effects of microgravity on tissue and organ development, as well as alterations in human physiology and fertility in long term space travels. Our immediate goal is to understand how polymerized microtubules form striated patterns under normal gravity but not under microgravity as has been reported by Tabony and co-workers (Papaseit et. al., PNAS, 2000, 97, 8364). We have reproduced most of the observations by Tabony et. al. under normal gravity conditions. The formation of the patterns is strongly influenced by a transient, convective flow, induced by an initial temperature gradient. Similar patterns can be reproduced by shearing polymerizing tubulin samples, suggesting strongly interplay between flow alignment and the continuous growth of microtubules and microtubule bundles following the rapid initial assembly of tubulin. Thus the physical mechanism of tubulin pattern formation is most likely attributable to coupling between gravity and convection, which profoundly affects all levels of biological and physiological functions in space travel. The objective of our ongoing investigation is threefold: 1, to precisely define the role of gravity on pattern formation; 2, to decouple the effects of convection and gravity using the Variable Magnetic Force (VMF) apparatus; and 3, to design a strategy for simulated gravity or certain desirable effects of convection in space travel environment. Convection is one of the fundamental mechanisms for heat transport and thermal equilibration. It works in a container of fluid, as well as within the tissues and organs in our body. Since the warmer liquid tends to be lighter, thermal inhomogeneity efficiently dissipates on earth, facilitated by the coupling between convection and gravity. Our body is well adapted during the course of evolution on earth to utilize this physical mechanism, but the possible adverse effects due to lack of this mechanism in space travel are poorly recognized. Among the 45 risks listed in the NASA Bioastronautics Roadmap for human space exploration, our study may be relevant to Risk 4. Renal stone formation, and Risk 40. Maintain thermal balance. Current countermeasures listed include enhanced fluid uptake and selection of thermal transporting fluids to address the two risks, respectively. A better strategy to facilitate flow and fluid transport both within the human body and in the space environment is crucial for space travel. Our study uniquely addresses this important issue using a protein matrix with essential biological functions including cell division and tissue development. The ground based research program we conduct may also be relevant to Risk 2-Impaired fractural healing, Risk 6-Diminished cardiac and vascular function, and a yet to be addressed issue-reproduction in long term space exploration.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This ground based research program defines the physical and biochemical mechanisms of tubulin assembly, which plays important functional roles in cell division and tissue development. The relevant mechanism bears resemblance to and suggests potential side effects of certain human fertility treatment. The results of this study may suggest useful measures to alleviate the side effects. This research program explores the effects of a high strength, high gradient magnetic field on the tubulin samples. This approach demonstrates a way of manipulating biological samples that is underutilized at the present.
Task Progress:	The PI and co-PI each attended two NASA meetings within the past 10 months. Both PI and co-PI attended the half-day NASA meeting in November, 2004, for principal investigators working in the field of space biology. The co-PI was invited to give a talk in the 35th COSPAR Scientific Assembly in Paris, July, 2004. The PI was invited to present a talk in the 2005 NASA cell science conference in Feb, 2005. In addition, the PI presented results of the NASA sponsored program in the New England Complex Fluids Workshop, a quaterly regional meeting, in Dec, 2004, and in March, 2005. The PI also gave invited seminars on the NASA sponsored research program in Washington Univ (Oct, 04), Rhode Island Colleage (Nov, 04) and UMass (Dec, 04).
Bibliography Type:	Description: (Last Updated: 11/01/2007)
Articles in Peer-reviewed Journals	Valles, J.M., Maris, H.J., Seidel, G.M., Tang, J.X., and Yao, W. "Magnetic Levitation Based Martian and Lunar Gravity Simulator" Journal of Advances in Space Research , Apr-2005
Dissertations and Theses	Mihalusova, M. "Preliminary Analysis of Microtubule Pattern Formation" Senior Thesis, Division of Biomedical Research, Brown University May-2004